ECE 285 Homework 1: Generating Chest X-Rays with a Variational Autoencoder

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1 Method

Data. We use the PNEUMONIA class of the Kaggle chest X-ray pneumonia dataset, which has about $3\times$ more images than the NORMAL class. Its 8 official `val` images are merged into `train`, giving 3,883 training images. The 390 PNEUMONIA `test` images are held out for validation loss and as the FID reference set. Each film is converted to grayscale and resized to 128×128 without cropping so that the full lung fields are kept, with intensities scaled to $[0, 1]$.

Model. The encoder is five stride-2 4×4 convolutions (32–64–128–256–512 channels, BatchNorm, LeakyReLU 0.2) down to a $512 \times 4 \times 4$ feature map, followed by two linear heads for the mean $\boldsymbol{\mu}$ and log-variance $\log \boldsymbol{\sigma}^2$ of $q_\phi(\mathbf{z} \mid \mathbf{x})$ with $\mathbf{z} \in \mathbb{R}^{128}$. The decoder mirrors it with transposed convolutions and outputs per-pixel Bernoulli logits (8.7 M parameters in total). We minimise

$$\mathcal{L}(\mathbf{x}) = \underbrace{-\mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} \log p_\theta(\mathbf{x} \mid \mathbf{z})}_{\text{BCE, summed over pixels}} + \beta_t \, \mathrm{KL}(q_\phi(\mathbf{z} \mid \mathbf{x}) \,\|\, \mathcal{N}(\mathbf{0}, \mathbf{I})) \,, \tag{1}$$

with $\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\epsilon}$. The KL weight β_t is annealed linearly from 0.05 to 1 over the first 20 epochs to avoid early posterior collapse; for all later epochs (and for every reported number) $\beta = 1$, i.e. $\mathcal{L}$ is the true negative ELBO. New images are generated by decoding $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and taking the Bernoulli mean $\sigma(\text{logits})$.

Hyperparameters. Table 1 lists the settings, all set in `config.py`. Training takes 4.3 minutes on a laptop RTX 5060.

Evaluation protocol. FID and IS use PyTorch-Ignite's `FID` and `InceptionScore` metrics as in the Ignite GAN-evaluation tutorial: images in $[0, 1]$ are repeated to three channels and bilinearly resized to 299×299 before torchvision's InceptionV3. Ignite's FID needs equally sized real and candidate sets, so each candidate set has 390 images, compared with the 390 held-out test X-rays over 5 random draws (mean $\pm$ std). This pipeline gives small absolute FIDs (the tutorial reports $\approx$0.016 for a DCGAN on CelebA), so real training images are scored against the same reference as a floor.

Code. https://github.com/devapi016/ece285-deep-generative-ai/tree/hw1-submission/hw1

2 Results

Training dynamics. Figure 1 shows the loss curves. While β is small the KL term starts high and then drops quickly as β rises. It settles at $\approx$53 nats per image without collapsing, so the latent code stays informative. The final negative ELBO is 9,460 nats on train and 9,520 on held-out data. The held-out reconstruction term flattens after about 120 epochs while the training term keeps falling, a mild overfit of $\approx 0.6\%$. The held-out loss never rises (its minimum is at epoch 197), so we keep the final checkpoint.

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Table 1: Hyperparameters.

Image size	$1 \times 128 \times 128$
Latent dim	128
Optimiser	Adam, lr 10^{-3}
LR schedule	cosine $\rightarrow 10^{-5}$
Batch size	64
Epochs	200
KL weight β	$0.05 \rightarrow 1$ over 20 ep.
Likelihood	Bernoulli (BCE)
Seed	0

Table 2: Image quality against the 390 held-out test X-rays (5 draws of 390 images; lower FID and higher IS are better).

Image set	FID ↓	IS ↑
Real train images (floor)	0.008 ± 0.001	2.13 ± 0.07
VAE reconstructions	0.206 ± 0.007	1.87 ± 0.03
VAE samples (prior)	$\mathbf{0.163 \pm 0.006}$	$\mathbf{1.74 \pm 0.03}$
Real test images	–	1.96

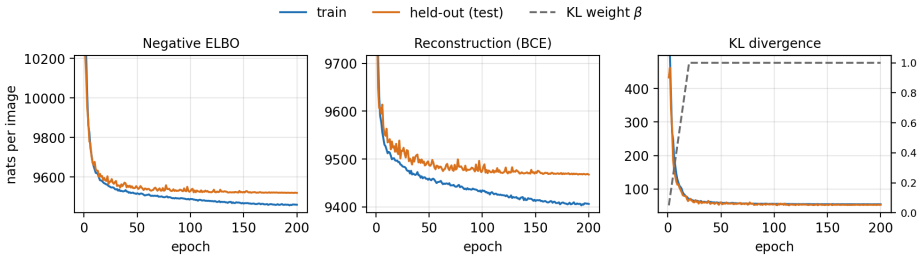
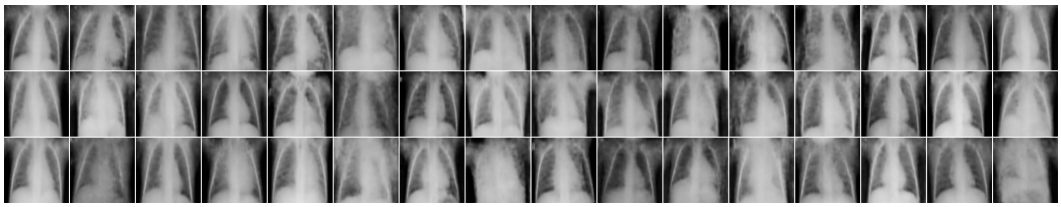


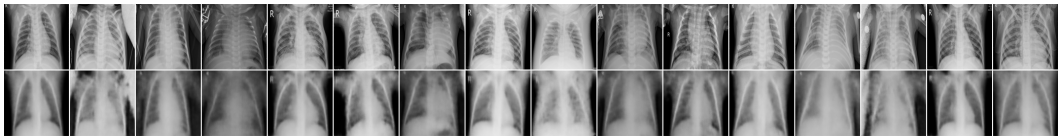
Figure 1: Per-image training curves (epoch 1 is off-scale). Held-out losses are computed with $\beta = 1$ after every epoch. The dashed line is the KL annealing schedule.

Generated images. The samples in Figure 2a show the key anatomy: clavicles, a rib cage that narrows toward the apices, the cardiac silhouette and mediastinum, the diaphragm domes and the dark film border. They also reproduce the class's variety: patchy and diffuse airspace opacities of different severity, sometimes one-sided, together with rotated or narrow (paediatric) chests. Reconstructions (Figure 2b) keep pose and lung opacity but lose fine rib edges, text and devices.

Sample quality. VAE samples reach an FID of 0.163 (Table 2) versus 0.008 for real training images, so a large gap to real data remains. The gap is mostly due to *blur*, not to a mismatch between prior and posterior: reconstructions of real images score no better (0.206) than samples, so the decoder itself cannot produce high-frequency texture. This is the expected failure mode of a pixel-wise factorised likelihood, which averages over the uncertain positions of fine rib and vessel edges. IS is barely informative here, because ImageNet classes do not describe X-rays: even real test images only reach 1.96, versus 1.74 for samples. A perceptual or adversarial loss would be the natural next step.



(a) 48 random samples decoded from $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.



(b) Held-out test originals (top) and their reconstructions from the posterior mean (bottom).

Figure 2: Generated and reconstructed chest X-rays (128×128).